A VCG-Style Mechanism for the Grading Problem

Target paper: Gans & Kominers, *What Does a Grade Mean? Informativeness and Strategic Manipulation of Grading Systems* (NBER w35183).

What "the problem" is, in mechanism-design terms. Latent performance is additive, $P_{ic} = a_i - d_c + \varepsilon_{ic}$, where a_i is student ability (the object the labor market wants) and d_c is a *manipulable nuisance*: students manipulate it by **course-shopping** (choosing low-d_c courses), instructors manipulate it by **leniency** (lowering d_c to attract enrollment). A scalar grade cannot separate a_i from d_c (their Thm 4.5). The paper's fix — *eigengrades* — nets out an estimate of d_c recovered from the transcript network, achieving incentive-neutrality $\Omega(\mathcal{R}) = 0$.

The request was a **VCG-like mechanism** for this problem. The honest finding (after adversarial checking with Codex/gpt-5.5 and numerical Monte-Carlo, see §5) is that there are **two distinct routes** to the same efficient outcome, and they map onto the two halves of the VCG idea:

Route	VCG analogue	What it does	Where it lives
Eigengrade as Groves-form correction	the *informational* side: net out a nuisance using others' data	changes the **measurement** so the report is d-invariant	§2 (students)
Pivotal (leave-one-out) eigengrade	the *Clarke-pivot* made literal (uses only θ_{-i})	self-influence-free correction	§3 (students)
Pigouvian/Groves transfer	the *transfer* side: price the externality	keeps raw grades, **prices** the leniency externality	§4 (instructors)

The strongest *new, constructive, verified* contribution is §4: a genuine Groves/Pigouvian transfer mechanism that implements professional grading in **strictly dominant strategies** while *keeping raw grades*, as an alternative to changing the signal. §2–§3 sharpen the paper's student-side incentive result and give the precise sense in which eigengrades are — and are *not* — a Groves mechanism.

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1. The two strategic margins and the efficient target

- **Students** choose courses to maximize pedagogical value v_{ic} plus the labor-market value of the resulting report, $m_i(c; \mathcal{R}) = \mathbb{E}[\hat{a}_i(Y_i^{\mathcal{R}}) \mid a, i \text{ chooses } c]$. Efficiency = course choice driven by v_{ic} alone, i.e. $\Omega(\mathcal{R}) = 0$.
- **Instructors** choose mean grades $\bar{G}_c$; the inefficiency is the *competitive enrollment* term $\beta\eta_{\mathcal{R}}(1-1/m)/\xi$ in the grade-inflation equilibrium (their Prop 8.3). Efficiency = the professional profile $\bar{G}_c = \bar{G}_0 + \alpha/\xi$.

Both inefficiencies are driven by the same object — the manipulable course effect d_c loading on the reported signal. VCG offers two levers on such a nuisance: **remove it from the signal** (information design — §2–§3) or **charge for it** (transfers — §4).

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2. Eigengrades as a Groves-form correction (and where the analogy stops)

A linear full-matrix score is $T_i(P) = \sum_{j,c} L_{i,jc} P_{jc}$. The paper's Prop 6.3 characterizes the valid adjustments. Restated as a clean quotient-space fact (verified):

> **Proposition 2.1 (annihilator characterization).** Let $D = \{\mathbf{1}_n h^\top : h \in \mathbb{R}^m\}$ be the course-shift subspace of grade matrices. The linear scores invariant to every additive course shift $P \mapsto P - \mathbf{1}_n h^\top$ are **exactly** the annihilator $D^\perp$, i.e. the linear functionals on $\mathbb{R}^{n\times m}/D$ — equivalently the coefficient arrays with $\sum_j L_{i,jc} = 0$ for every course c (their eq. 6). Among these, the ones returning $a_i - \bar{a}$ for all (a, d) are exactly those that *also* satisfy $\sum_c L_{i,jc} = \mathbf{1}\{j = i\} - 1/n$ (their eq. 7).

The Groves reading — and its precise limits. Split $T_i(P) = \underbrace{\sum_c L_{i,ic} P_{ic}}_{\text{own grades}} + \underbrace{H_i(P_{-i})}_{\text{others' grades}}$. Condition (6) says: *when a course is shifted, the change in the own-grade part is exactly cancelled by the change in the reference term H_i built from others' data.* That is structurally the Clarke-pivot idea — cancel a contaminating common component using information the agent does not control. Two honest caveats (confirmed by Codex against Green–Laffont/Holmström):

1. **It is an analogy, not an equivalence.** There is no money, no allocation rule, and no quasilinear payoff identity $v_i(k^*) + t_i = \text{surplus} + h_i(\theta_{-i})$ here. Condition (7) is an *identification* normalization (it pins the target functional $a_i - \bar{a}$), **not** a transfer normalization. "Eigengrades *are* Groves mechanisms" is too strong as a *formal* claim.
2. **Their Prop 6.2** ("no d-invariant score built from a student's own grades is non-trivial") is the *analogue* of the Green–Laffont/Holmström impossibility that own information cannot simultaneously identify the object and strip the nuisance — you are forced to use cross-agent information (the pivot). Again: same idea, different category (linear identification vs. dominant-strategy incentive compatibility).

The defensible theorem-environment statement is Proposition 2.1 plus: *these corrections are Groves-like only in the limited sense that they use cross-agent information to cancel an external nuisance component.* The literal VCG mechanism is §4.

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3. The pivotal (leave-one-out) eigengrade: the Clarke pivot made literal

To make the pivot structure *literal* — a correction that is a function of θ_{-i} **alone** — define, for each student i, the leave-one-**student**-out course-effect estimate. Delete student i and all her edges; on the remaining graph G_{-i} compute the graph-Laplacian estimate and read off the course block $\hat{d}^{(-i)}$. The **pivotal eigengrade** for a focal course slot c is

$$Y_i(c) = P_{ic} + \hat{d}_c^{(-i)}, \qquad \hat{d}^{(-i)} \text{ depends on } \{P_{jc} : j \neq i\} \text{ only.}$$

> **Proposition 3.1 (first-moment incentive-neutrality, exact).** If G_{-i} is connected, then under the additive noisy model with mean-zero shocks independent of course choice,
>
> $$m_i(c; \mathcal{R}) = \mathbb{E}[Y_i(c) \mid a, d, i \text{ chooses } c] = a_i - \kappa_{-i} \quad \text{for every feasible } c,$$
>
> where κ_{-i} does **not** depend on i's course choice. Hence $b_i(c)$ is constant in c and $\Omega(\mathcal{R}) = 0$ **exactly**. Under a true external anchor, $\kappa_{-i} = 0$ and $m_i(c) = a_i$ (conditionally

unbiased *and* choice-invariant).

Three things this clarifies — two of them sharpen the paper, one is a caveat the paper hedged correctly.

- **(Sharpening.)** For the *linear* eigengrade, first-moment neutrality is **exact**, not merely "approximate as noise shrinks." The reason: $\mathbb{E}[\hat{d}^{(-i)}] = d - \kappa_{-i}\mathbf{1}$ is a single common shift for *any* connected graph, because $\ker L_{-i} = \operatorname{span}\{(\mathbf{1}, \mathbf{1})\}$ is graph-independent. The **standard** (non-leave-out) eigengrade is *also* exactly neutral in the unconditional mean — Monte-Carlo confirms a null effect (z = 0.27, §5). So the paper's "approximate under noise" is conservative for the cardinal-linear case.
- **(What leave-out genuinely buys.)** The standard estimator lets student i's own shock enter $\hat{d}_c$ with weight $-1/N_c$ (class-size leverage): $\partial\hat{d}_c/\partial\varepsilon_{ic} = -1/N_c \neq 0$. The leave-out estimator sets this to **exactly zero** — the defining Clarke-pivot property (no self-influence; a student cannot grade herself up through the adjustment). This is a *structural* virtue, not a first-moment one.
- **(Honest caveat — the genuine residual channel.)** Exactness is **in expectation**, not ex-post in every realized sample (Codex's $n = 3$ counterexample: realized *other-student* shocks make any single transcript sample-dependent). The remaining strategic channel under a **Bayesian** market is **second-moment**: course choice changes the *precision* of the report (via the Laplacian leverage / class size, their Thm 6.15 spectrum), so a shrinkage market's posterior mean can still depend on c. Closing this requires a *report-only* (no-shrinkage) market — which the paper assumes — or a precision-equalized report. This is the precise location of the paper's "approximate" hedge.

  Proposition 3.2 (the price of the pivot — bias/efficiency tradeoff). Under zero-sum leave-out normalization, $\kappa_{-i} = \bar{a}_{-i} = (A - a_i)/(n-1)$, so the cross-student ranking statistic carries bias $(m_i - m_j) - (a_i - a_j) = (a_i - a_j)/(n-1) = O(1/n)$. It is removed exactly by an external anchor, or corrected by the rescaling $\tilde{Y}_i = \frac{n-1}{n} Y_i$ (which preserves within-student course-invariance). Without an anchor, absolute ability is unidentified (the global additive constant), so only centered ability is recoverable.

This is the familiar VCG/jackknife trade: making the pivot use *only* others' data (exact non-manipulability) costs an $O(1/n)$ centering bias, recovered by anchoring — exactly the role the paper's external anchor already plays for ordinal data (their §6.5).

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4. A genuine VCG/Groves mechanism for instructor grade inflation

This is the literal mechanism (transfers + dominant-strategy efficiency), and the main constructive deliverable. It targets the instructor game of their §8.

Environment (their §8). Each course c is an instructor choosing mean grade $\bar{G}_c \in \mathbb{R}$. Logit enrollment shares $s_c(\bar{G}) = e^{\eta \bar{G}_c} / \sum_k e^{\eta \bar{G}_k}$ ($\eta = \gamma\rho_{\mathcal{R}} \geq 0$ the report-mediated pass-through; $\eta = 0$ under exact eigengrades). Intrinsic payoff

$$U_c(\bar{G}) = \alpha\bar{G}_c + \beta \log s_c(\bar{G}) - \tfrac{\xi}{2}(\bar{G}_c - \bar{G}_0)^2, \qquad \alpha, \beta \geq 0,\ \xi > 0.$$

Their Prop 8.3: unique symmetric equilibrium $\bar{G}^{\mathrm{NE}} = \bar{G}_0 + \dfrac{\alpha + \beta\eta(1 - 1/m)}{\xi}$. The term $\beta\eta(1 - 1/m)/\xi$ is the **competitive-enrollment component of grade inflation** — pure businessstealing, since shares sum to one. The planner values only the non-rent-seeking part, $W(\bar G) = \sum_k[\alpha\bar G_k - \frac{\xi}{2}(\bar G_k-\bar G_0)^2]$, maximized at the **professional profile** $\bar G_c = \bar G_0 + \alpha/\xi$.

The mechanism. Keep raw grades; pay instructor c the transfer

$$\boxed{t_c(\bar G) = -\beta\log s_c(\bar G) + h_c(\bar G_{-c})}$$

where h_c depends only on other instructors' grades.

> **Theorem 4.1 (Groves-form, dominant-strategy efficiency).** *(verified — §5)* (i) **Groves payoff identity:** $U_c+t_c = \big[\alpha\bar G_c - \frac{\xi}{2}(\bar G_c-\bar G_0)^2\big] + h_c(\bar G_{-c})$ — the $\beta\log s_c$ terms cancel exactly. Equivalently, with the full Groves term $t_c^G = \sum_{k\neq c} f_k(\bar G_k) - \beta\log s_c + h'_c$, one gets $U_c + t_c^G = W(\bar G) + h'_c(\bar G_{-c})$, so each instructor's payoff equals social welfare up to an others-only constant; the minimal transfer is $-\beta\log s_c +$ (function of others). (ii) **Dominant strategy:** the residual objective is strictly concave with maximizer $\bar G_c^* = \bar G_0 + \alpha/\xi$ **independent of** $\bar G_{-c}$**,** η**, and** β. So the professional profile is the unique strictly dominant-strategy equilibrium, *even when* $\eta > 0$ (raw grades still used). The competitive-inflation term is eliminated by **pricing**, not by changing the signal. (iii) **Equivalence to eigengrades:** the implemented outcome $\bar G_0 + \alpha/\xi$ equals the $\eta = 0$ eigengrade equilibrium of Cor. 8.4. Pricing the externality and removing it from the signal are two routes to the same grading. (Equivalence is at the grading-outcome level only; raw grades still carry d_c for any downstream channel outside W.)

Why dominance (and the contrast). After cancellation the objective is separable, so the optimum is opponent-independent — *strictly dominant*, not merely Nash. The natural "internalize-others" alternative $\tilde t_c = \beta\sum_{k\neq c}\log s_k$ gives FOC $\alpha - \xi(\bar G_c - \bar G_0) + \beta\eta(1-ms_c) = 0$, which equals the target *only* at the symmetric profile $s_c = 1/m$ — i.e. Nash, not dominant. (Both confirmed numerically.)

> **Proposition 4.2 (budget — the standard VCG obstruction, honest).** *(corrected after verification — my first draft wrongly claimed no-subsidy/balance.)* The Clarke-pivot freedom h_c **cannot** be chosen to make $t_c \le 0$ for all $\bar G_c$ on the unbounded domain (since $\log s_c \to -\infty$ as $\bar G_c \to -\infty$, the transfer turns into a subsidy for $\bar G_c < \bar G_c^*$). Moreover **exact budget balance** $\sum_c t_c = 0$ **is impossible** under smooth dominant-strategy implementation whenever $m \ge 2, \beta > 0, \eta > 0$: balance would force the full mixed derivative $\partial_{1\cdots m}\sum_c U_c$ to vanish at the target, but $\partial_{1\cdots m}(\beta\sum_c\log s_c) = \beta(-1)^m m!\,\eta^m\prod_c s_c = \beta(-1)^m m!\,\eta^m/m^m \neq 0$. This is the textbook VCG budget tension. **Remedy:** an **AGV / expected-externality** transfer restores budget balance *in expectation*, at the cost of moving from dominant-strategy to **Bayes-Nash** implementation (requires a prior over instructor cost types). On a **bounded** grade interval $[\underline G, \overline G]$ the target is the projection of $\bar G_0 + \alpha/\xi$, and a finite per-instructor entry fee makes the scheme ex-post individually rational.

Reading. §4 is the genuine VCG-style result: it is *Groves-form* (Theorem 4.1) and prices a real externality to implement efficiency in dominant strategies; it inherits VCG's classic no-budget-balance limitation (Prop 4.2). Calling it strictly "VCG/Clarke" is justified only once it is embedded in a direct-revelation environment (instructors report cost types $\theta_c = (\alpha_c, \xi_c, \bar G_{0,c})$, the planner picks the welfare- maximizing standard, and h_c is the Clarke pivot = others' welfare at c's worst-case absence); in the action-game form above it is most precisely a **Groves/Pigouvian corrective transfer**.

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5. Verification status (Codex/gpt-5.5 high-effort + Monte-Carlo; all findings triaged)

Claim	Method	Verdict
Prop 2.1 (annihilator / eqs 6–7)	Codex re-derivation	**Confirmed** both directions.
"Eigengrades *are* Groves"	Codex adversarial	**Downgraded to analogy** — no money/allocation/payoff identity. Stated as such.
Prop 3.1 (first-moment $\Omega = 0$ exact)	Codex + MC (z=0.27 std; anchor matches a_i)	**Confirmed in expectation.** My "ex-post in every sample" claim was **false** (Codex $n = 3$ counterexample) — corrected to "in expectation."
Standard eigengrade already first-moment exact	MC + Codex A3	**Confirmed** — my original premise (leave-out needed for first-moment exactness) was **wrong**; corrected to "leave-out removes *self-influence*."
Prop 3.2 ($O(1/n)$ bias, $\kappa_{-i} = \bar{a}_{-i}$, rescale fix)	Codex	**Confirmed**, with exact leading term.
Thm 4.1 (i)–(iii) (Groves-form, dominant-strategy)	Codex line-by-line + MC	**Confirmed** (BR $= \bar{G}_0 + \alpha/\xi$ for all opponents; Nash-contrast verified).
Thm 4.1(iv) no-subsidy / budget balance	Codex	**FALSE — removed.** Replaced by honest Prop 4.2 (subsidy region; balance impossible for $m \geq 2, \beta, \eta > 0$; AGV restores balance only in Bayes-Nash).

Artifacts: prompts in `output/codex_prompts/`, Codex transcripts in `output/codex_explorations/`, Monte-Carlo scripts `output/leaveout_montecarlo.py` and `output/vcg_instructor_check.py`.

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6. Bottom line

- The paper's eigengrade is best understood as the **informational half** of VCG — a Groves-*form* correction that cancels the course-effect nuisance using cross-student data (§2). The **leave-one-out** version makes the pivot literal (self-influence-free) and shows first-moment incentive-neutrality is *exact*, not approximate, with the genuine residual being a second-moment precision channel (§3).
- The **transfer half** of VCG gives a genuinely new, dominant-strategy mechanism for the instructor grade-inflation game (§4): a Pigouvian/Groves tax $-\beta \log s_c$ on enrollment-stealing implements professional grading *while keeping raw grades* — an alternative to eigengrades

that reaches the same outcome by **pricing** the externality instead of **removing it from the signal**. It carries the standard VCG budget-balance limitation, repairable in expectation via AGV.